

Meeting Report

AI/Automation Platform for Purchase & Sales Workflow

Discovery, Requirements & Process Mapping

Client Domain	Metal Scrap Trading (Copper, Aluminum, Brass, HMS)
Prepared For	Internal Stakeholders & Project Team
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Meeting Report: AI/Automation System for Purchase & Sales Workflow

Meeting Overview

- **Topic:** Discovery and requirement gathering for an AI/automation platform to streamline purchase contracts, sales contracts, invoice processing, and shipment tracking for a metal scrap trading business
- **Client Business:** Metal scrap trading (primarily copper, aluminum, brass, HMS) — brokering between suppliers (Saudi Arabia, UAE, etc.) and end customers (China, India, etc.)
- **Key Stakeholders:**
 - **Karan Desai (KD)** — Business lead, demonstrating current manual processes
 - **Shivendrasinh Patankar (SP)** — Solution architect from the implementation team
 - **Pornima Kaduba Gaikwad (PG)** — Business analyst / functional consultant
 - **Shubham Adepawar (SA)** — Accounting team member (explained SAP posting)
 - **Rampravesh Shah (RS)** — Additional stakeholder
- **Objective:** Understand the as-is purchase and sales workflows, capture requirements, and design an automation platform to replace manual Excel-based processes, integrate with SAP, and reduce manual effort/errors

1. Business Background

1.1 Trade Model

- The company deals in **non-ferrous metal scrap** (copper, aluminum, brass, HMS) and operates in a **brokerage + proprietary trading model**.
- **Deal flow distribution:**
 - **~90% of deals** are routed through brokers (broker provides both supplier and customer)
 - **~10% of deals** are self-sourced — the company has its own supplier or customer
 - Of self-sourced deals, roughly **20–30%** are through the company's own customers, the rest through brokers
- **Volume:** 5 to 20 deals per day, each involving multiple containers (typically 25 MT per 20-ft container for copper)
- **Geography:** Suppliers primarily in Saudi Arabia, UAE, Israel; Customers in China and India

1.2 Commodity Codes (Standard Abbreviations)

Code	Commodity
CU	Copper
AL / ALU	Aluminum
CUZNS	Brass
MIX	Mixed metal scrap
HMS	Heavy Melting Steel
TIP	Other (catch-all)

1.3 Key Pricing Concepts

- **LME (London Metal Exchange):** Benchmark price
- **LLME:** Lowest LME price
- **Provisional invoice:** Price not yet fixed — pay 80% of calculated value
- **Final invoice:** Price fixed — pay 100%
- **Performa invoice:** Advance / clean invoice without weight slip — requires CEO approval
- **Balance invoice:** Final settlement after provisional
- **Fixed price:** Price locked at contract stage
- **Discount factor (percentage):** e.g., 97% of LME = 3% discount
- **LME Hedging:** Complex risk-mitigation mechanism (out of scope for Phase 1)
- **X-yard vs CNF:** "X-yard" means all freight costs are on the buyer after goods leave supplier's yard; "CNF" includes freight

1.4 Documents Required by Destination

- **China:** Invoice, Packing List, Certificate of Origin, Chemical Analysis, Mill Test Certificate (if required)
- **India:** Invoice, Packing List, Certificate of Origin, Freight Certificate, Form 6, Form 9, Mill Test Certificate, Chemical Analysis

2. As-Is Process — Purchase Workflow

2.1 Deal Origination

1. Deal comes via email from broker/supplier (or directly from the HOD/sir)

2. Reference codes:

- **Broker's reference number**

- **LLME / LME percentage** (e.g., 97% of LME)
 - **Quantity** (e.g., 125 MT = 5 × 25 MT containers)
 - **Loading date** (typically 125 hours from deal confirmation)
3. The team creates an entry in the "**Loading Sheet**" (master Excel file) which is the brain of the operation
- **Batch number format**: 1 + last 2 digits of financial year + 2 digits of company + sequence number (e.g., 17026-642)
 - The loading sheet is verified by 2 people daily to ensure data integrity

2.2 Purchase Contract Creation

- **Two types of purchase contracts:**

1. Supplier provides the contract → team uses it
2. Company creates the contract based on deal terms
 - **Standard template** in Excel with fixed fields; only the **yellow-highlighted fields** change per contract:
 - Date, address, description, origin
 - Quantity, percentage
 - Payment terms, delivery terms
 - **Highlight color coding** in the loading sheet:
 - **Yellow** = not fixed (provisional)
 - **Blue** = fixed
 - **Container marking**: 20-ft or 40-ft size mentioned in email/document
 - **Standard commodity per container**: 25 MT (20-ft) for copper; varies for aluminum
 - The contract is converted to PDF, **physically signed & stamped** by HOD (DocuSign NOT applicable for scrap trading)
 - Stored in a dated folder structure

2.3 Exception Cases in Purchase

- **Broker gives both supplier and customer** (most common) — company only provides money and earns margin
- **Company has own supplier, no customer yet** — find a customer, then create contract
- **Company has own customer, no supplier yet** — find a supplier
- **Brokerage deals** (~80–95%): Standard 80% advance, 20% balance
- **B2B deals** (mainly with **Necro Metals**): Full advance payment on provisional; profit-sharing (50-50, 60-40, etc.); expenses shared

- **UAE restriction:** License does not permit buying from UAE and selling within/out of UAE → these are booked under Singapore entity (code 3010) instead of UAE entity (2000)

2.4 Invoice Receipt & Processing

- Invoice arrives via email
- The system should determine **provisional vs final** by checking:
 - Explicit label on invoice
 - Or hedging data / fixed price in loading sheet (columns Q, R, S, T)
- **Quantity rules:**
 - Always **net quantity** (gross weight minus container tare weight)
 - **20-ft container** = ~25 MT for copper
 - **40-ft container** = ~25 MT for aluminum (UAE); varies by region
- **Rounding tolerance:**
 - Maximum \$1 difference is acceptable
 - Anything above \$1 must be flagged; person creating can self-approve
 - >\$10 difference → HOD approval required
- **Advance payment logic:**
 - Provisional = 80% advance
 - Final = 100%
 - Balance = remaining amount
 - Performa invoice = CEO approval required (no weight slip/draft bill)
- **Validation** before posting:
 - Arithmetic check: quantity × price
 - DPR created with same amount
 - Batch number matches
 - LME rate matches purchase contract
 - Quantity matches weight slip
 - LME rate + batch number cross-check via loading sheet (lowest & highest of hedging day)

2.5 SAP Posting (Current Process)

- Documents uploaded to **DMS (Document Management System)** by commercial team
- Accounting team opens each PDF, manually populates Excel, then posts to **SAP** via:
 - **GTM module** → T-code **MIRO** (Invoice Verification)

- **GRN (Goods Receipt Note)** posted separately
 - **SAP fields populated manually:**
- Reference Key 1 & 2
- Invoice number, invoice date, posting date
- Baseline date, due date
- House Bank (company's bank)
- DMS document number (for audit traceability)
- Business Area = **1070** (common code for scrap business)
- Assignment field = Invoice number
- Header Text = Batch number
- Vendor code (party code)
 - Final submit posts debit/credit entries
 - **Current pain point:** 2 hours per day spent just opening documents

2.6 OBL (Original Bill of Lading) Handling

- OBL is the "passport" of a container — required to release the container via bank/shipping line
- Once OBL arrives:
 - Cross-check container number
 - Merge with invoice, sign with HOD
 - Rename to **PR (Purchase Receipt)** and re-upload to DMS
 - Sales invoice is raised simultaneously, creating sales TC number
 - **Weight discrepancy handling:**
 - Compare OBL weight vs invoice weight vs loading sheet
 - If invoice weight > OBL weight → raise **debit note** to supplier
 - If OBL weight > invoice weight → dispute with supplier, raise **credit note** to customer if shortage is significant

2.7 Date Validation (MSME / Indian Law)

- **Current rule:** No strict date criteria
- **Proposed rule:** Invoices older than 3 months should not be accepted
- **Indian MSME law:** If payment terms are 45 days, invoice must come at least 45 days before due date; otherwise interest liability arises
- Company to confirm tolerance and approval matrix for backdated invoices

3. As-Is Process — Sales Workflow

3.1 Sales Contract Types

1. **Fixed price** — Price locked at contract stage
2. **LME %-based** — Price = LME × % (final price fixed later based on hedging)
3. **On LME** — Indexed to LME price

3.2 Sales Contract Process

- Two territories: **India** and **China** (different document requirements)
 - **Customer name abbreviation**: First 3 letters of customer name (e.g., "DONG" for DongA → "DON")
- For purchase: First 2 letters of each word
 - Company to provide master list of customer abbreviations
 - System auto-generates for new customers
 - **Reference fields that change per contract (yellow-highlighted)**:
 - Customer ID
 - Commodity name
 - Location
 - Port
 - LME data, percentage
 - Payment terms
 - Delivery (X-shot, FOB, etc.)
 - Broker name + commission (~\$25 per MT)
 - **Clause #3 (Price) & #4 (LME hedging)** are mutually exclusive — only one applies
 - **Sales to China** may have different document requirements
 - **Payment terms**:
 - 100% CAD / DP through bank
 - TT / Telex (Telex release)
 - **Sales contract template** is the same as purchase, just different color coding (purple for sales fields)

3.3 Sales Invoice Process

- Sales invoice raised when container is loaded
- **LME price source**: West Metal website (no public API available)

- **Solution:** User manually enters 3-month LME price for the day
- **Advance / Proforma invoice:**
 - 10% advance is standard
 - Calculation: $\text{LME price} \times \% \times 10\%$
 - Advance is deducted from final invoice
 - Deviation from 10% advance must be flagged
- **Commission:** $\$25/\text{MT} \times \text{quantity}$
- **OBL process for sales:**
 - Customer releases container at destination port
 - Risk of customer releasing without paying → follow-up needed

3.4 Sales Invoice Amendments / Edge Cases

- **Price amendment:**
 - Quality problem → quality penalty
 - Price amendment (rare)
 - Discount factor (e.g., 97% of LME)
 - Tolerance rules: Price must be exact; quantity tolerance ~3.3% is industry standard
- **Contract amendment:**
 - Above threshold: Quantity, price → HOD/CEO approval via chatbot workflow
 - Basic change (provisional → fixed price): No approval needed
 - Below threshold: Self-approval
 - Approval matrix (DOA) to be configurable in front-end
 - Audit trail mandatory for all amendments

3.5 Sales Pricing Mechanism Variants

- **Fixed price** (locked at contract)
- **Index price** (linked to LME)
- **Percentage of LME** (e.g., 97%)
- **3-month LME** (averaged over 3 months before ETD/ETA)

3.6 Performance Invoice

- 10% advance against provisional
- $\text{LME} \times \% \times 10\%$ calculation

- Bank guarantee / BGs not used

4. Operations & Tracking

4.1 Daily Operations

- Constant follow-up on documents, vendor KYC, container receipt verification
- **Reports required:**
 - How many countries shipped to
 - Average timeline
 - Customer/supplier-wise revenue
 - Custom reports on demand
- **One person spends 3–4 hours every morning** manually tracking 300–500 containers via shipping line websites

4.2 Container Tracking

- Currently manual via shipping line websites
- Need web scraping / API integration to extract:
 - Get out / get out empty date
 - Port departure / arrival
 - ETA
 - IC arrival
- Previous project experience shows third-party websites can be used for this
- **Alert triggers:**
 - ETA changes
 - Status changes
 - Daily dashboard for operations & approvals

4.3 Segregation in Operations

Pornima proposed four workstreams:

1. **Sales** workflow
2. **Purchase** workflow
3. **Shipment / Cargo tracking**
4. **Operational reporting**

4.4 Contract / Invoice Numbering

- **Loading sheet columns:** One of the most important identifier columns used across the workflow
- **Contract #, PO #, DPR #** are unique per container (not generic)
- **TC and DPR** are run via SAP separately for each invoice
- TC is created first, then PO copied, then DPR runs

5. AI / Automation Platform — Requirements

5.1 Core Features

1. Email integration:

- Read incoming mails (subject line keywords: "please find deal confirmation", "please share purchase contract", "please share sales contract")
- Create contract in the system
- Reply to the same email thread (initially via human-approved draft)

2. Master data management:

- Supplier master
- Customer master (with addresses, bank details)
- Vendor/customer name abbreviations
- Broker master
- Commodity master
- Container size master

3. Vendor portal (already in demo):

- Drag-and-drop invoice upload
- Email sync option

4. Document generation:

- Purchase contract (PDF)
- Sales contract (PDF)
- Invoice (PDF, physical signature still required)
- Performa invoice
- Bank cover letter (auto-fill PDF)

5. Two-way channel ingestion:

- Direct email sync
- Manual drag-and-drop upload

5.2 Validation & Approval Rules

- **Tolerance / rounding:**

- Max \$1 difference auto-acceptable
- \$1–\$10 difference self-approval
- >\$10 → HOD approval

- **Price amendment:**

- Exact match required (no tolerance)
- Quantity tolerance ~3.3% (industry standard)

- **Weight validation:**

- Invoice weight vs OBL weight
- Invoice weight vs weight slip
- Tare weight of container

- **Advance % validation:**

- 10% is standard
- Above 80% advance → HOD approval
- Performa invoice without weight slip → CEO approval

- **Date validation:**

- Invoice > 3 months old → reject / HOD approval
- Future-dated invoices → reject

- **DOA (Delegation of Authority) matrix** — configurable in front-end

- **Routing:**

- If approver absent/leave → escalate to next level
- Sequential or parallel approval configurable
- TAT per approval level (24 hrs / 48 hrs)

- **Major vs Minor changes:**

- Major → HOD approval
- Minor → self-approval
- Must be clearly defined for AI to decide

5.3 UI / UX Requirements

- **Layout:** System-generated document on left, editable right side (similar to cement business demo)

- **Contract section** on the left of the screen
- **Small chatbot on the right** for changes/inputs
- **Three hubs** to keep organized:
 - **Contract hub**
 - **Invoice hub**
 - **Sales hub**
- **Visual consistency** across all screens
- **Standard terminology display** (right side always editable)
- **Drop-down overrides** for change of wording (3-month, LME, other)
- **Mandatory reason** for wording changes
- **Chatbot for amendments:** Captures user intent, formats formal note, sends to approval queue with change summary, impact, and DOA-based routing
- **Loading sheet interface:** Designed like an Excel sheet for easy adoption

5.4 Audit & Compliance

- **Audit trail** on every change/override
- **HOD signature** required for original physical contracts
- Email replies must be **system-generated** with disclaimer ("do not reply to this mail")
- **Multi-user tracking:** Track who created, edited, approved each document

5.5 Integration Points

- **SAP:**
 - TC (Purchase Order) creation
 - DPR (goods receipt/invoice) creation
 - GRN (Goods Receipt Note) creation
 - MIRO (Invoice Verification) via GTM module
 - F-53 / F-58 for payment clearing
 - Multiple company codes: 2000 (UAE), 3010 (Singapore)
 - Same SAP instance used
 - **Email platforms:** Outlook/Microsoft (company is on Microsoft, not Google)
 - **OCR for invoice extraction:** PDFs from DMS to Excel to SAP
 - **API integration:**
 - OCR → Excel → SAP API
 - Web scraping for shipping line tracking

- West Metal (LME) — no API, manual entry

5.6 AI / ML Capabilities

- **OCR** for invoice data extraction (handles image-based PDFs)
- **Read and interpret email threads** — detect final email vs trail, understand references
- **Web scraping** for container tracking
- **Decision-making logic:**
 - Provisional vs final detection
 - Tolerance checks
 - Approval routing
 - Major vs minor change classification
 - **Custom reports** generated automatically (daily/weekly, CXO view)
 - **Validation score** with **weighted parameters** (not simple average) — to be defined

5.7 Data Points the System Must Capture

For each deal:

- Reference number (broker's)
- Batch number (auto-generated, format: I7026-642)
- Contract number (auto-generated)
- TC number, PO number, DPR number
- LME hedging data (hedging date, LME price, final price)
- Hedging range (lowest & highest on hedging day)
- Container number, size (20/40)
- Quantity (net)
- Commodity code & description
- Origin / load port
- Discharge port
- Customer name + abbreviation
- Supplier name + abbreviation
- Broker name + commission
- Payment terms, delivery terms
- Invoice number, date
- Weight slip details
- OBL number

- Status tracking (loaded, in transit, delivered, etc.)
- Advance payment %
- Final price

6. Approval & Workflow Configuration

6.1 Approval Matrix (Configurable)

- **Purchase contract changes:**

- > 5 quantity / > 20 ton (or similar thresholds) → RM1 + RM2 approval

- **Container amendments:**

- Increase in quantity → contract amendment workflow

- Fixed price change to indexed → basic practice, no approval

- **Override / exception cases:**

- All exceptions captured with reason in dropdown

- Bot generates formal note

- Sent to approver queue

6.2 Rejection Logic

- If invoice amount exceeds calculation (hedging price × qty) → reject
- Supplier must revise with correct value
- If calculation > invoice (advance > invoice) → simple netting in vendor ledger
- If invoice > advance (final > provisional) → reverse DPR if standing in ledger > 2–3 days

6.3 B2B Workflow

- Full advance payment on provisional
- Profit-sharing (50-50, 60-40, 65-35, etc.) based on offer
- Expenses shared with B2B partner
- Loss borne by partner
- Separate workflow from brokerage deals
- Identified via "B2B" tag in loading sheet
- Filter / dashboard by deal type
- Currently: One person (Pornima's team) works on B2B differently

7. Gap Items / Open Questions (To Be Confirmed)

#	Item	Owner	Status
1	Maximum age for invoice acceptance (proposed: 3 months)	Karan / Management	TBD
2	Approval matrix for backdated invoices	Management / Audit	TBD
3	MSME / Indian law impact on E-invoicing	Senior/auditing team	TBD
4	Tolerance percentages for quantity / quality	Karan	TBD
5	Full list of small checks performed by accounting team	Karan (Shubham)	Pending
6	Full list of validation parameters (with weights)	Karan's team	Pending
7	Complete commodity list (plant codes)	Already partially shared	Pending
8	Master list of customer name abbreviations	Karan (pre-go-live)	Pending
9	New template for Loading Sheet (rev 6+)	Karan	Will share next week
10	SAP access for Singapore entity	IT team	TBD
11	Weight slip variations across suppliers (separate session)	Karan	TBD
12	DOA matrix details (HOD, CEO, etc.)	Management	TBD
13	Reports template (current standard reports)	Karan	Will share

#	Item	Owner	Status
14	Quality penalty / premium calculation logic	Karan	Discussed partially
15	X-yard vs CNF handling for specific suppliers	Karan	TBD

8. Pain Points & Expected Benefits

8.1 Current Pain Points

- **Manual Excel-based loading sheet** — prone to errors, requires 2-person verification
- **VBA scripts** for SAP posting — fragile, slow (5 min per document)
- **Email parsing** of long threads is manual
- **2 hours/day** spent just opening documents
- **3–4 hours/day** spent on container tracking (1 person, 300–500 containers)
- **Approval delays** due to long email chains and manual handoffs
- **PDFs of invoices** are image-based → hard to extract data
- **Communication via group email** ("promotion of DXP") — replies get lost in thread
- **Weight slip interpretation** is highly variable per supplier
- **Cross-checking 6–7 fields per DPR** takes 5 minutes each

8.2 Expected Benefits

- Reduce manual effort by **70–80%** in invoice posting
- **Auto-classify** provisional vs final, advance %
- **Auto-route** approvals based on DOA matrix
- **Single source of truth** for all contract/invoice data
- **Real-time visibility** for HOD, CXO
- **Audit-ready** with full trail
- **Faster** document turnaround
- **Drone-level tracking** with proactive alerts

9. Project Status & Timeline (As Discussed)

9.1 Project Status

- **Project initiation & governance:** 100% complete ■
- **Finalizing business requirements (AS-IS):** 25% → now ~30% (after these sessions)
- **Solution design & development:** Pending
- **UAT sign-off:** Target last week of October
- **Project handover (go-live):** Target **13 November**

9.2 Milestones

Date	Milestone
19 October	UAT Start
End October	UAT sign-off
13 November	Go-live (project handover)

9.3 Next Steps

- **Monday/Tuesday:** BRD delivery to Karan
- **Wednesday/Thursday:** Technical Design Document (TDD)
- **Thursday:** Quotation (pricing) delivery
- **Next Tuesday:** Sales process session
- **Internal MOM:** Pornima to document all open items and decisions pending

10. Key Terminology Glossary

Term	Definition
LME	London Metal Exchange — benchmark price for metals
LLME	Lowest LME — minimum price benchmark
DPR	Down Payment Request — advance payment record in SAP
TC	Purchase Order (Trading Contract) in SAP
GRN	Goods Receipt Note
OBL / OBL	Original Bill of Lading — container's "passport"

Term	Definition
PR	Purchase Receipt (post-OBL)
DMS	Document Management System (SAP-based)
MIRO	SAP T-code for invoice verification
GTM	Goods/Trade Module in SAP
X-yard	Buyer pays all freight after goods leave supplier's yard
CNF	Cost & Freight
CAD / DP	Cash Against Documents / Documents against Payment
TT / Telex	Telex Transfer (wire transfer)
FOB	Free On Board
CIF	Cost, Insurance & Freight
HMS	Heavy Melting Steel
MT / MTS	Metric Ton(s)
KYC	Know Your Customer
DOA	Delegation of Authority
B2B	Business-to-business (profit sharing with supplier like Necro)
P&L;	Profit & Loss
ETS / ETA	Estimated Time of Shipment / Arrival
ETD	Estimated Time of Departure
Batch No	Format: I + last 2 digits of FY + company code + sequence

11. Sign-off

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Reviewed by: Karan Desai

Distribution: All meeting attendees

Next Review: BRD delivery (Monday/Tuesday)

End of Report